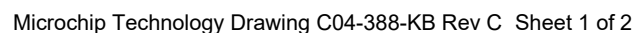
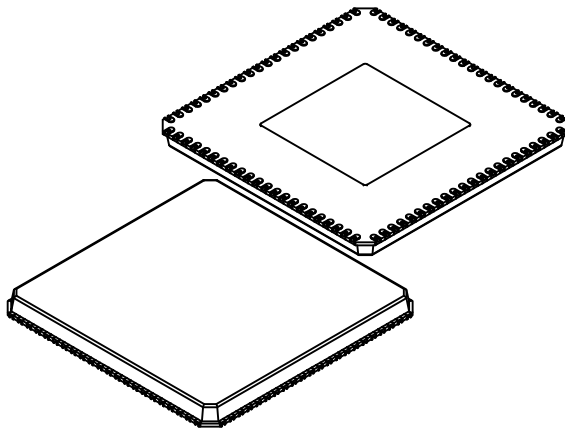


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

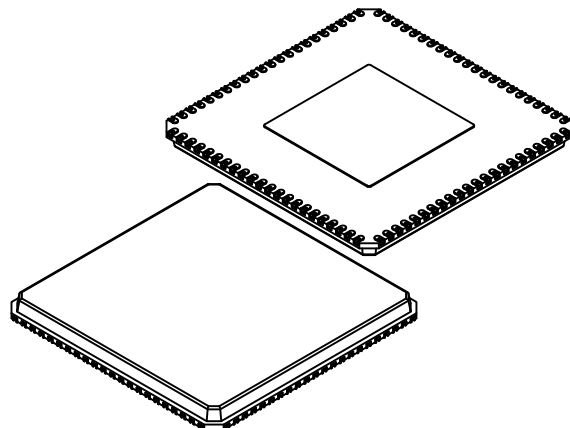


88-Lead Very Thin Plastic Quad Flat, No Lead Package (KB) 12x12x0.9 mm Body [VQFN] Punch Singulated With 6.0x6.0 mm Exposed Pad and Dimpled Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Original Configuration



Alternate Configuration

		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		88		
Pitch	e		0.50 BSC		
Overall Height	A		0.80	0.85	0.90
Standoff	A1		0.00	0.02	0.05
Terminal Thickness	A3		0.20 REF		
Overall Length	D		12.00 BSC		
Mold Cap Length	D1		11.75 BSC		
Exposed Pad Length	D2		5.90	6.00	6.10
Overall Width	E		12.00 BSC		
Mold Cap Width	E1		11.75 BSC		
Exposed Pad Width	E2		5.90	6.00	6.10
Terminal Width	b		0.18	0.25	0.30
Terminal Length	L		0.30	0.40	0.50
Terminal-to-Exposed-Pad	K		2.60 REF		
Mold Draft Angle	θ		—	—	12°

		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Package Corner Chamfer	CH		—	—	0.60
Terminal Dimple Height	A5		0.09 REF		
Terminal Dimple Width	b1		0.10	0.15	0.20
Terminal Dimple Length	L1		0.05	0.15	0.25
Package Corner Chamfer	CH		—	—	0.60

ALTERNATE CONFIGURATION

Overall Height	A	0.80	0.85	0.90
Terminal Thickness	A3	0.20 REF		
Base Thickness	A4	0.30 REF		
Mold Cap Length	D1	11.71 BSC		
Mold Cap Width	E1	11.71 BSC		

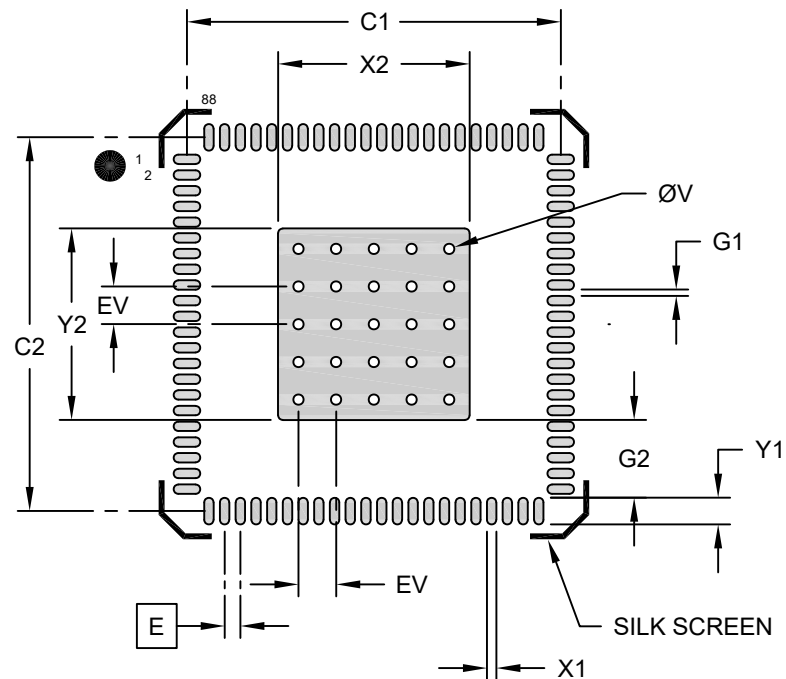
Terminal Dimple Height	A5	0.05	0.10	0.15
Terminal Dimple Width	b1	0.10	0.15	0.20
Terminal Dimple Length	L1	0.05	0.15	0.25

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is punch singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

**88-Lead Very Thin Plastic Quad Flat, No Lead Package (KB) 12x12x0.9 mm Body [VQFN]
Punch Singulated With 6.0x6.0 mm Exposed Pad and Dimpled Wettable Flanks**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

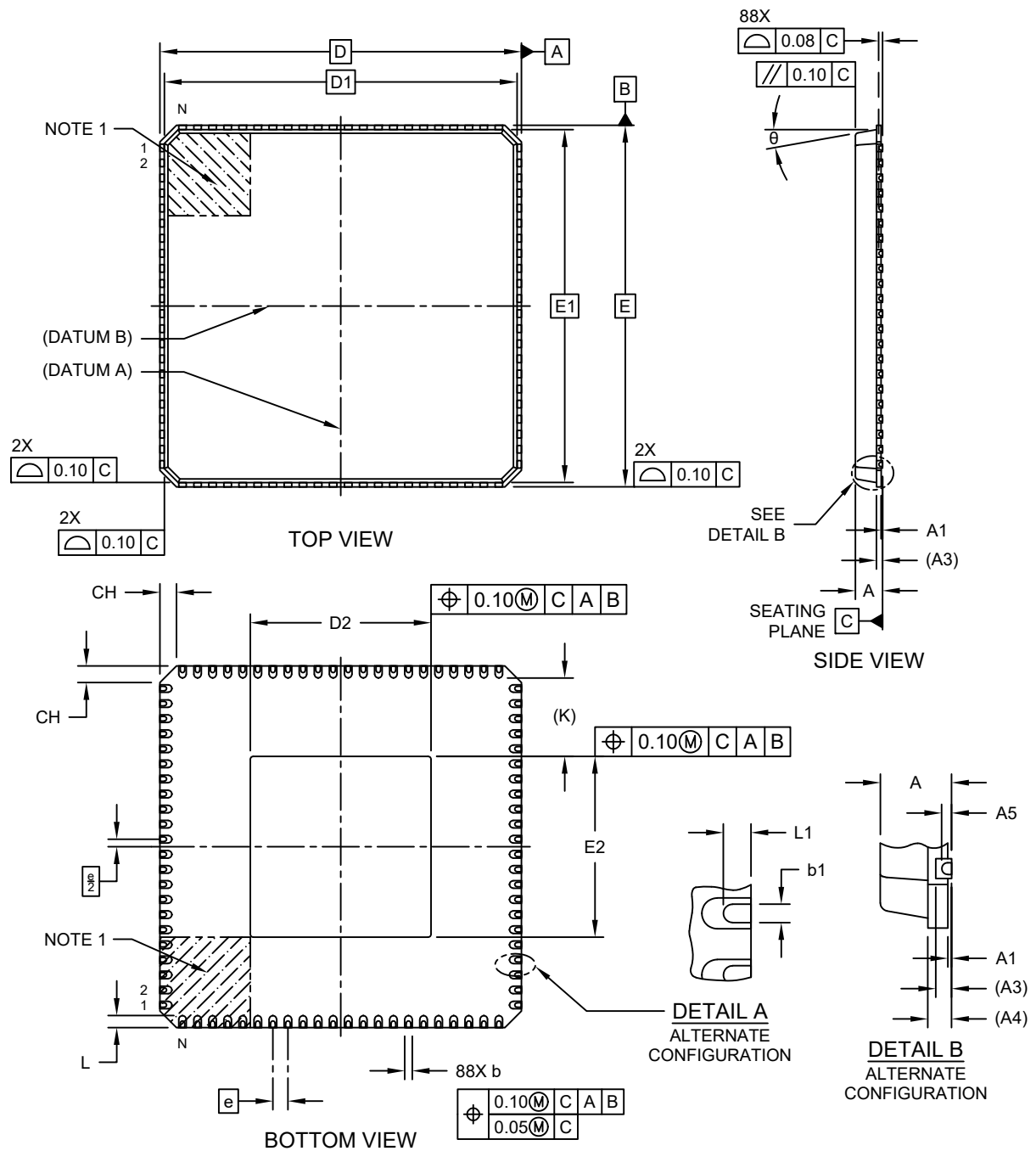
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Optional Center Pad Width	X2			6.90
Optional Center Pad Length	Y2			6.90
Contact Pad Spacing	C1		11.90	
Contact Pad Spacing	C2		11.90	
Contact Pad Width (X88)	X1			0.30
Contact Pad Length (X88)	Y1			0.80
Contact Pad to Pad (X84)	G1	0.20		
Contact Pad to Center Pad (X88)	G2		2.48	
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

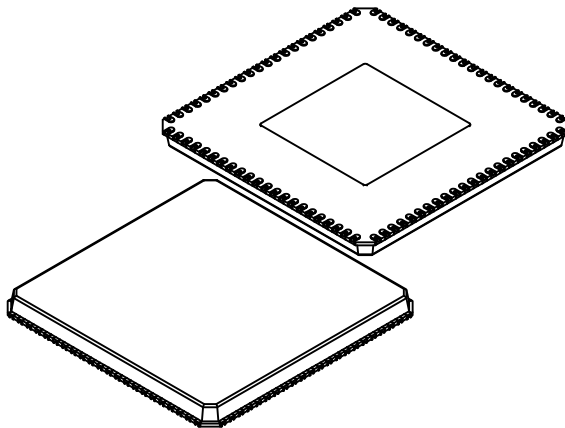
**88-Lead Very Thin Plastic Quad Flat, No Lead Package (KBX) 12x12x0.9 mm Body [VQFN]
Punch Singulated With 6.0x6.0 mm Exposed Pad and Dimpled Wettable Flanks**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

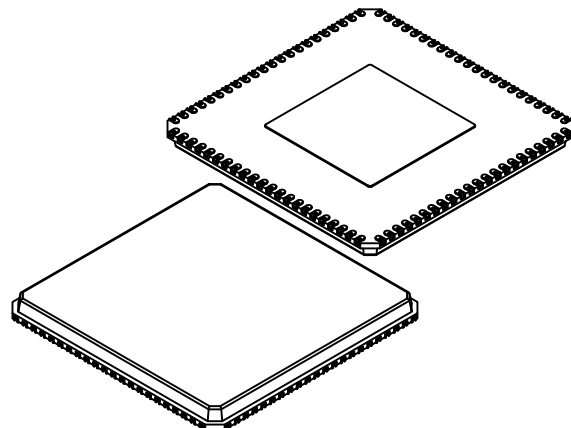


88-Lead Very Thin Plastic Quad Flat, No Lead Package (KBX) 12x12x0.9 mm Body [VQFN] Punch Singulated With 6.0x6.0 mm Exposed Pad and Dimpled Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Original Configuration



Alternate Configuration

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	88		
Pitch	e	0.50 BSC		
Overall Height	A	0.80	0.85	0.90
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.20 REF		
Overall Length	D	12.00 BSC		
Mold Cap Length	D1	11.75 BSC		
Exposed Pad Length	D2	5.90	6.00	6.10
Overall Width	E	12.00 BSC		
Mold Cap Width	E1	11.75 BSC		
Exposed Pad Width	E2	5.90	6.00	6.10
Terminal Width	b	0.18	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	2.60 REF		
Mold Draft Angle	θ	—	—	12°

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Package Corner Chamfer	CH	—	—	0.60
Terminal Dimple Height	A5	0.09 REF		
Terminal Dimple Width	b1	0.10	0.15	0.20
Terminal Dimple Length	L1	0.05	0.15	0.25
Package Corner Chamfer	CH	—	—	0.60

ALTERNATE CONFIGURATION

Overall Height	A	0.80	0.85	0.90
Terminal Thickness	A3	0.20 REF		
Base Thickness	A4	0.30 REF		
Mold Cap Length	D1	11.71 BSC		
Mold Cap Width	E1	11.71 BSC		

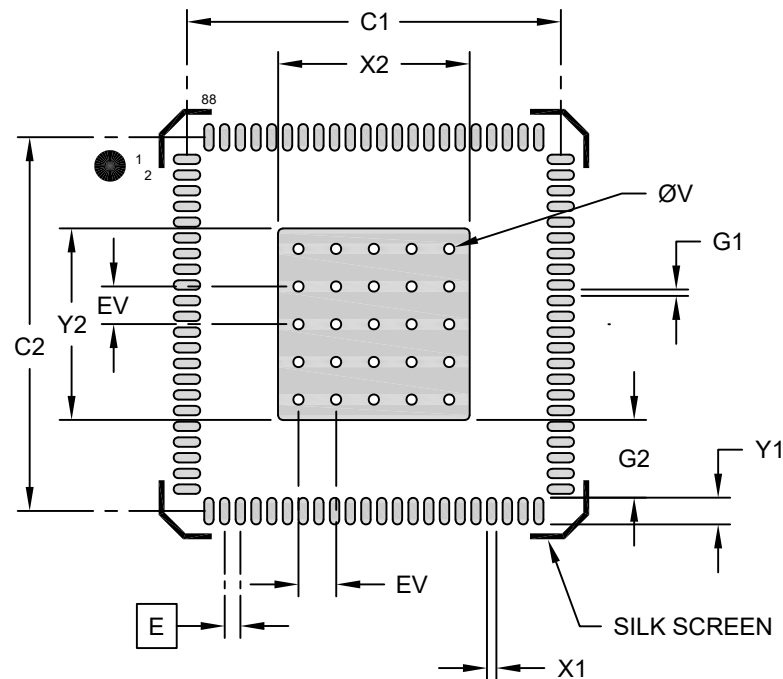
Terminal Dimple Height	A5	0.05	0.10	0.15
Terminal Dimple Width	b1	0.10	0.15	0.20
Terminal Dimple Length	L1	0.05	0.15	0.25

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is punch singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

**88-Lead Very Thin Plastic Quad Flat, No Lead Package (KBX) 12x12x0.9 mm Body [VQFN]
Punch Singulated With 6.0x6.0 mm Exposed Pad and Dimpled Wettable Flanks**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Optional Center Pad Width	X2			6.90
Optional Center Pad Length	Y2			6.90
Contact Pad Spacing	C1		11.90	
Contact Pad Spacing	C2		11.90	
Contact Pad Width (X88)	X1			0.30
Contact Pad Length (X88)	Y1			0.80
Contact Pad to Pad (X84)	G1	0.20		
Contact Pad to Center Pad (X88)	G2		2.48	
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process